Digital Logic and Computer Design Laboratory

3rd Semester, Academic Year 2026

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Week#8 Program Number:8

WRITE A VERILOG PROGRAM TO CONSTRUCT COMPARTOR AND PREFIX ADDER. GENERATE THE VVP OUTPUT AND SIMULATION WAVEFORM USING GTKWAVE. VERIFY THE OUTPUT AND WAVEFORM WITH THE TRUTH TABLE

I. Verilog Code Screenshot
II. Verilog VVP Output Screen Shot
III. GTKWAVE Screenshot
IV. Output Table (Truth Table) to be completed and included

```
λ  vvp comparator.vvp
VCD info: dumpfile comparator.vcd opened for output.
                   0 a=0000 b=0000 | greater=0 equal=1 less=0
                  20 a=0101 b=0011 | greater=1 equal=0 less=0
                  40 a=0010 b=0110 | greater=0 equal=0 less=1
                  60 a=1000 b=1000 | greater=0 equal=1 less=0
                  80 a=1111 b=0001 | greater=1 equal=0 less=0
                 100 a=0100 b=1100 | greater=0 equal=0 less=1
                 120 a=1010 b=1010 | greater=0 equal=1 less=0
comparator_tb.v:20: $finish called at 140 (1s)
```

```
λ  vvp prefix_adder.vvp
VCD info: dumpfile prefix_adder.vcd opened for output.
                   0 a=0000 b=0000 cin=0 | sum=0000 cout=0
                  20 a=0101 b=0011 cin=0 | sum=1000 cout=0
                  40 a=1111 b=0001 cin=0 | sum=0000 cout=1
                  60 a=1010 b=0101 cin=1 | sum=0000 cout=1
                  80 a=1111 b=1111 cin=0 | sum=1110 cout=1
                 100 a=1111 b=1111 cin=1 | sum=1111 cout=1
prefix_adder_tb.v:21: $finish called at 120 (1s)
```

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```
1  module comparator(
2      input [3:0] a, b,
3      output greater, equal, less
4  );
5      assign greater = (a > b);
6      assign equal = (a == b);
7      assign less = (a < b);
8  endmodule
9
```

```
module comparator_tb;
    reg [3:0] a, b;
    wire greater, equal, less;

    comparator uut(a, b, greater, equal, less);

    initial begin
        $dumpfile("comparator.vcd");
        $dumpvars(0, comparator_tb);

        $monitor($time, " a=%b b=%b | greater=%b equal=%b less=%b", a, b, greater, equal, less);

        a = 4'b0000; b = 4'b0000;
        #20 a = 4'b0101; b = 4'b0011;
        #20 a = 4'b0010; b = 4'b0110;
        #20 a = 4'b1000; b = 4'b1000;
        #20 a = 4'b1111; b = 4'b0001;
        #20 a = 4'b0100; b = 4'b1100;
        #20 a = 4'b1010; b = 4'b1010;
        #20 $finish;
    end
endmodule
```

```
module prefix_adder(
    input [3:0] a, b,
    input cin,
    output [3:0] sum,
    output cout
);
    wire [3:0] p, g, c;

    assign p = a ^ b;
    assign g = a & b;

    assign c[0] = cin;
    assign c[1] = g[0] | (p[0] & c[0]);
    assign c[2] = g[1] | (p[1] & g[0]) | (p[1] & p[0] & c[0]);
    assign c[3] = g[2] | (p[2] & g[1]) | (p[2] & p[1] & g[0]) | (p[2] & p[1] & p[0] & c[0]);
    assign cout = g[3] | (p[3] & g[2]) | (p[3] & p[2] & g[1]) | (p[3] & p[2] & p[1] & g[0]) | (p[3] & p[2] & p[1] & p[0] & c[0]);

    assign sum = p ^ c;
endmodule
```

```verilog
module prefix_adder_tb;
    reg [3:0] a, b;
    reg cin;
    wire [3:0] sum;
    wire cout;

    prefix_adder uut(a, b, cin, sum, cout);

    initial begin
        $dumpfile("prefix_adder.vcd");
        $dumpvars(0, prefix_adder_tb);

        $monitor($time, " a=%b b=%b cin=%b | sum=%b cout=%b", a, b, cin, sum, cout);

        a = 4'b0000; b = 4'b0000; cin = 1'b0;
        #20 a = 4'b0101; b = 4'b0011; cin = 1'b0;
        #20 a = 4'b1111; b = 4'b0001; cin = 1'b0;
        #20 a = 4'b1010; b = 4'b0101; cin = 1'b1;
        #20 a = 4'b1111; b = 4'b1111; cin = 1'b0;
        #20 a = 4'b1111; b = 4'b1111; cin = 1'b1;
        #20 $finish;
    end
endmodule
```



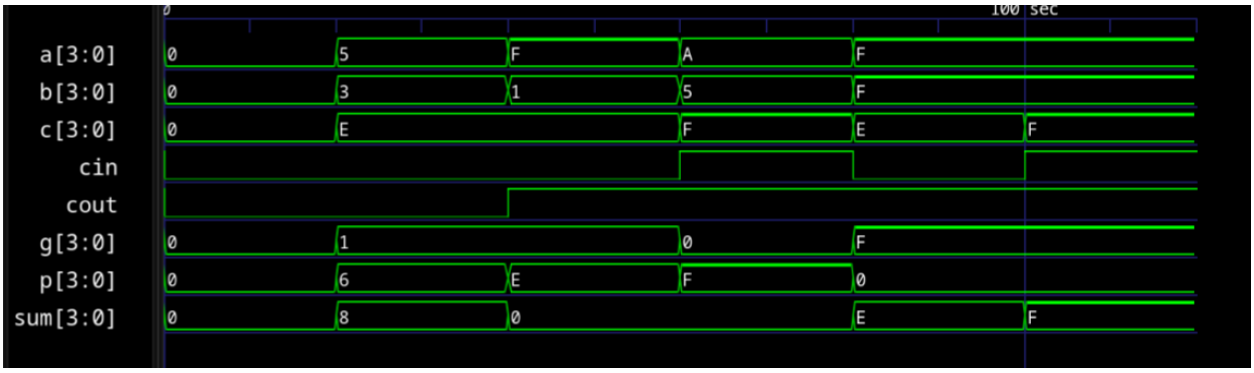


Table 1: 4-Bit Magnitude Comparator Verification Table

Test Case	A [3:0] (Binary)	A (Decimal)	B [3:0] (Binary)	B (Decimal)	Greater (A > B)	Equal (A = B)	Less (A < B)
1	0000	0	0000	0	0	1	0
2	0101	5	0011	3	1	0	0
3	0010	2	0110	6	0	0	1
4	1000	8	1000	8	0	1	0
5	1111	15	0001	1	1	0	0
6	0100	4	1100	12	0	0	1
7	1010	10	1010	10	0	1	0

Table 2: 4-Bit Prefix / Carry-Lookahead Adder Verification Table

Test Case	A [3:0] (Binary)	A (Dec)	B [3:0] (Binary)	B (Dec)	Cin	Sum [3:0] (Binary)	Sum (Dec)	Cout
1	0000	0	0000	0	0	0000	0	0
2	0101	5	0011	3	0	1000	8	0
3	1111	15	0001	1	0	0000	0	1
4	1010	10	0101	5	1	0000	0	1
5	1111	15	1111	15	0	1110	14	1
6	1111	15	1111	15	1	1111	15	1

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